

Binomial series — 6.4

$\frac{1}{1-x}$, e^x , $\sin(x)$, $\cos(x)$ have series we know

Today: consider a power series for functions of the form $f(x) = (1+x)^\alpha$, e.g. $f(x) = \frac{1}{\sqrt{1+x}}$
 $\alpha = -\frac{1}{2}$

Recall if f has a power series representation, it is the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

$$f(x) = (1+x)^\alpha \Rightarrow f(0) = (1+0)^\alpha = 1^\alpha = 1$$

$$f'(x) = \alpha(1+x)^{\alpha-1} \Rightarrow f'(0) = \alpha(1)^{\alpha-1} = \alpha$$

$$f''(x) = \alpha(\alpha-1)(1+x)^{\alpha-2} \Rightarrow f''(0) = \alpha(\alpha-1)$$

$$f'''(x) = \alpha(\alpha-1)(\alpha-2)(1+x)^{\alpha-3} \Rightarrow f'''(0) = \alpha(\alpha-1)(\alpha-2)$$

$$f^{(n)}(x) = \alpha(\alpha-1)(\alpha-2) \dots (\alpha-n+1)(1+x)^{\alpha-n}$$

$$f^{(n)}(0) = \alpha(\alpha-1)(\alpha-2) \dots (\alpha-n+1) \quad " = \frac{\alpha!}{(\alpha-n)!} "$$

multiplying n total things

Coefficient in the Taylor series for x^n

$$\text{is then } \frac{f^{(n)}(0)}{n!} = \frac{\alpha(\alpha-1)(\alpha-2) \dots (\alpha-n+1)}{n!} \quad " = \frac{\alpha!}{(\alpha-n)! n!} "$$

$$" = \binom{\alpha}{n} "$$

We've seen $\frac{m!}{(m-n)!n!}$ before when studying probability. This is the number of ways of choosing n things from a collection of m things.

For example: $\star, \square, \circ, \triangle$, there are

$$\frac{4!}{(4-2)!2!}$$

ways to pick two of these shapes

$$\frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 2 \cdot 1} = 6$$

$\star \square$	$\square \circ$
$\star \circ$	$\square \triangle$
$\star \triangle$	$\circ \triangle$

Notation: $\frac{m!}{(m-n)!n!} = \binom{m}{n}$ "binomial coefficients"
say "m choose n"

We will write $\binom{\alpha}{n}$ to mean

$$\frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+2)(\alpha-n+1)}{n!}$$

← n total terms in numerator.

Taylor series for $(1+x)^\alpha$ is then

$$\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n.$$

$$\left[\begin{array}{l} \binom{\alpha}{0} = 1 \text{ for any } \alpha \\ \binom{\alpha}{1} = \alpha \text{ for any } \alpha \end{array} \right]$$

Example: $\alpha = -\frac{1}{2}$, $\frac{1}{\sqrt{1+x}} = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} x^n$ on $(-1, 1)$

$$= \binom{-\frac{1}{2}}{0} + \binom{-\frac{1}{2}}{1}x + \binom{-\frac{1}{2}}{2}x^2 + \binom{-\frac{1}{2}}{3}x^3 + \dots$$

$$= 1 - \frac{1}{2}x + \frac{\binom{-\frac{1}{2}}{2}(-\frac{3}{2})}{2!}x^2 + \frac{\binom{-\frac{1}{2}}{3}(-\frac{5}{2})}{3!}x^3 + \dots$$

$$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{15}{48}x^3 + \dots$$

When does $\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ converge? Ratio test!

$$\lim_{n \rightarrow \infty} \left| \frac{\binom{\alpha}{n+1} x^{n+1}}{\binom{\alpha}{n} x^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1)(\alpha-n)}{(n+1)!} x^{n+1}}{\frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1)}{n!} x^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1)(\alpha-n)}{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1)} \cdot \frac{n!}{(n+1)!} \cdot \frac{x^{n+1}}{x^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| (\alpha-n) \cdot \frac{1}{n+1} \cdot x \right| = |x| \lim_{n \rightarrow \infty} \left| \frac{\alpha-n}{n+1} \right| = |x|$$

Converges for $|x| < 1$ — radius of convergence is 1.

Example: Find a power series for $\sqrt[3]{1+x^2}$ and write out the first four nonzero terms.

This x^2 plugged into $(1+x)^{\frac{1}{3}} = \sum_{n=0}^{\infty} \binom{\frac{1}{3}}{n} x^n$

$$\text{so } (1+x^2)^{\frac{1}{3}} = \sum_{n=0}^{\infty} \binom{\frac{1}{3}}{n} x^{2n}$$

$$= 1 + \frac{1}{3} x^2 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!} x^4 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!} x^6 + \dots$$

$$= 1 + \frac{1}{3} x^2 - \frac{1}{9} x^4 + \frac{5}{81} x^6 + \dots$$

Example: Find a power series for $\sqrt{4+x}$.

Binomial-looking with $\alpha = \frac{1}{2}$

$$\sqrt{4+x} = \sqrt{4\left(1+\frac{x}{4}\right)} = 2\sqrt{1+\frac{x}{4}}$$

2 times an $\frac{x}{4}$ plugged into $\sqrt{1+x} = \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} x^n$

$$\sqrt{4+x} = 2\sqrt{1+\frac{x}{4}} = 2 \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} \left(\frac{x}{4}\right)^n = \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} \cdot \frac{2}{4^n} x^n.$$